Digital System Design

Module 2 — Sequential Logic Design, Counters, and Finite State Machines

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Comprehensive, exam-ready lecture notes covering Latches, Flip-Flops, Race-Around Condition, Master-Slave JK, Ripple and Synchronous Counters, Shift Registers, Mealy/Moore FSMs, and PRBS/LFSR Generators.

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1. Latches and Flip-Flops

In sequential logic circuits, outputs depend not only on current inputs but also on past history stored in memory elements.

1.1 Latches vs. Flip-Flops

- **Latch:** A level-triggered bistable storage device that responds continuously to inputs whenever the enable/clock signal is at an active level (transparent).
- **Flip-Flop:** An edge-triggered storage device that samples inputs and updates outputs only at a discrete transition (positive rising edge $\uparrow$ or negative falling edge $\downarrow$) of the clock signal.
- **Setup Time (t_s):** The minimum time for which the data input must remain stable *before* the arrival of the active clock edge.
- **Hold Time (t_h):** The minimum time for which the data input must remain stable *after* the active clock edge.
- **Metastability:** An unstable intermediate output voltage state occurring if setup or hold times are violated.

1.2 The S-R Latch and Forbidden State

An S-R (Set-Reset) latch constructed with cross-coupled NOR gates operates as follows:

- $S = 0, R = 0 \implies Q_{n+1} = Q_n$ (No change / Hold state).
- $S = 0, R = 1 \implies Q_{n+1} = 0, \overline{Q}_{n+1} = 1$ (Reset state).
- $S = 1, R = 0 \implies Q_{n+1} = 1, \overline{Q}_{n+1} = 0$ (Set state).
- $S = 1, R = 1 \implies Q = \overline{Q} = 0$ (**Invalid / Forbidden State**). When both inputs simultaneously drop from 1 to 0, the circuit enters a race condition, settling into an unpredictable state.

1.3 J-K Flip-Flop and the Race-Around Condition

The J-K flip-flop eliminates the invalid state of the S-R flip-flop by feeding back outputs Q and $\overline{Q}$ to the input AND gates:

$$S = J \cdot \overline{Q}, \qquad R = K \cdot Q \tag{1}$$

When $J = K = 1$, the flip-flop toggles its output on each active clock: $Q_{n+1} = \overline{Q}_n$.

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The Race-Around Condition in Level-Triggered JK Flip-Flops

Definition: In a level-triggered JK flip-flop, when $J = K = 1$ and the clock pulse width t_w is greater than the propagation delay t_{pd} of the flip-flop ($t_{pd} < t_w$), the output toggles repeatedly and uncontrollably back and forth ($0 \to 1 \to 0 \to 1 \dots$) during the single high clock interval. At the end of the clock pulse, the final state of Q is indeterminate.
Mathematical Condition for Race-Around:

$$t_{pd} < t_w < \frac{T}{2} \tag{2}$$

where T is the clock period and t_w is the pulse duration.
Methods to Prevent Race-Around:

a. **Edge Triggering:** Design the circuit to respond only to steep clock transitions ($t_w \approx 0$).

b. **Propagation Delay Control:** Choose components such that $t_{pd} > t_w$ (difficult and limits maximum operating frequency).

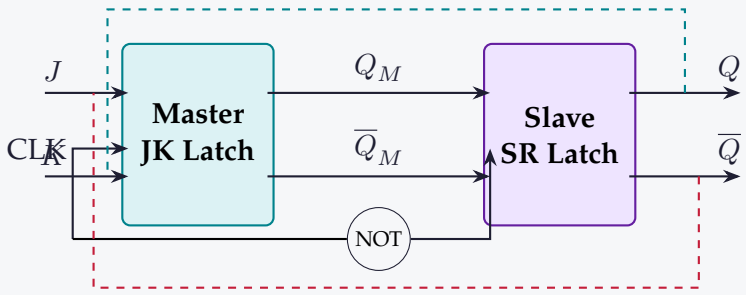
c. **Master-Slave JK Flip-Flop Configuration:** The universal and standard architecture.

1.4 Master-Slave JK Flip-Flop Architecture

The Master-Slave JK flip-flop consists of two cascaded latches clocked on complementary phases:

- **Master Latch:** Enabled when CLK $= 1$. It samples inputs J and K and updates its intermediate state (Q_M). The Slave latch remains disabled (isolated) because $\overline{\text{CLK}} = 0$.
- **Slave Latch:** When the clock falls ($\text{CLK} = 0 \implies \overline{\text{CLK}} = 1$), the Master is disabled, locking its state. The Slave is enabled and copies the state Q_M to the final external output Q.
- Because the output Q updates only on the clock falling edge when the Master is completely disabled, feedback cannot affect the Master during the same clock cycle, **completely eliminating race-around**.

Master-Slave JK Flip-Flop Functional Architecture



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1.5 Characteristic Equations and Excitation Tables

Type	Characteristic Equation	$Q_n \to Q_{n+1}$: $0 \to 0$	$0 \to 1$	$1 \to 0$	$1 \to 1$
S-R	$Q_{n+1} = S + \overline{R}Q_n$ $(SR = 0)$	$S = 0, R = X$	$S = 1, R = 0$	$S = 0, R = 1$	$S = X, R = 0$
J-K	$Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n$	$J = 0, K = X$	$J = 1, K = X$	$J = X, K = 1$	$J = X, K = 0$
D	$Q_{n+1} = D$	$D = 0$	$D = 1$	$D = 0$	$D = 1$
T	$Q_{n+1} = T \oplus Q_n$	$T = 0$	$T = 1$	$T = 1$	$T = 0$

2. Registers and Shift Registers

An n-bit register is a cascade of n flip-flops sharing a common clock, designed to store and shift multi-bit binary data words.

2.1 Shift Register Configurations

- **SISO (Serial-In Serial-Out):** Data enters one bit per clock cycle and exits serially. Requires n clock cycles to load and n clock cycles to read an n-bit word.
- **SIPO (Serial-In Parallel-Out):** Data enters serially; all n bits are accessible simultaneously on parallel outputs after n clock cycles.
- **PISO (Parallel-In Serial-Out):** Data is loaded into all flip-flops simultaneously in a single clock cycle using parallel load gates, then shifted out serially.
- **PIPO (Parallel-In Parallel-Out):** Parallel loading and parallel output reading simultaneously in 1 clock cycle.

2.2 Universal Shift Register (IC 74194)

A bidirectional 4-bit Universal Shift Register can perform all four data transfer operations controlled by two mode select inputs S_1, S_0:

$$S_1S_0 = 00 \implies \text{Hold}, \quad 01 \implies \text{Shift Right}, \quad 10 \implies \text{Shift Left}, \quad 11 \implies \text{Parallel Load} \tag{3}$$

3. Asynchronous (Ripple) and Synchronous Counters

A counter is a specialized sequential circuit that cycles through a prescribed sequence of binary states upon receiving clock pulses.

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3.1 Asynchronous (Ripple) Counters

In a ripple counter, the external clock is applied **only to the first flip-flop (LSB)**. Each subsequent flip-flop is clocked by the output of the preceding stage (Q or $\overline{Q}$).

- **Up-Counter vs. Down-Counter:**
  - Negative edge-triggered FF with clock connected to $Q_i \implies$ Up-counter.
  - Negative edge-triggered FF with clock connected to $\overline{Q}_i \implies$ Down-counter.
- **Mod-N Ripple Counter:** Formed by resetting all flip-flops using an active-low asynchronous clear ($\overline{\text{CLR}}$) driven by a NAND gate when the count reaches state N.
- **Disadvantages of Ripple Counters:**
  1. Accumulated delay: Total propagation delay is $t_{\text{total}} = n \times t_{pd}$.
  2. Maximum operating frequency is constrained: $f_{\max} \leq \frac{1}{n \cdot t_{pd}}$.
  3. Transient false counting states (glitches) appear on outputs during transition intervals.

3.2 Synchronous Counters

In a synchronous counter, the master clock is connected **in parallel to all flip-flops simultaneously**. State transitions occur at the exact same clock instant.

Systematic Synchronous Counter Design Procedure:

1. Determine number of flip-flops: $2^n \geq N$.
2. Construct the State Transition Diagram and State Table.
3. Substitute flip-flop excitation requirements (J_i, K_i or T_i or D_i) from the excitation table.
4. Minimize excitation expressions using Karnaugh maps.
5. Implement the logic circuit.

3.3 Ring Counter and Johnson (Twisted Ring) Counter

- **Ring Counter:** An n-bit shift register with true output Q_n fed back to serial input D_1. A single circulating '1' ($1000 \to 0100 \to 0010 \to 0001$).

$$\text{Number of unique states} = n, \qquad \text{Unused states} = 2^n - n \tag{4}$$

- **Johnson (Twisted Ring) Counter:** An n-bit shift register with inverted output $\overline{Q}_n$ fed back to D_1.

$$\text{Number of unique states} = 2n, \qquad \text{Unused states} = 2^n - 2n \tag{5}$$

  Requires only simple 2-input AND gates for complete glitch-free state decoding.

4. Finite State Machines (FSM)

Any synchronous sequential circuit can be modeled formally as a Finite State Machine consisting of combinational logic and state memory registers.

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4.1 Mealy vs. Moore FSM Models

Architectural Definitions of Mealy and Moore Machines

- **Moore Machine:** The output Z depends **only on the present state** S:

$$Z(t) = \lambda(S(t)) \tag{6}$$

  Outputs are synchronous with the clock and immune to input glitches.

- **Mealy Machine:** The output Z depends on **both the present state** S **and current inputs** X:

$$Z(t) = \lambda(S(t), X(t)) \tag{7}$$

  Outputs change asynchronously as soon as inputs change, enabling faster response and generally requiring fewer states (1 state fewer than Moore).

Block Diagrams: Moore Machine vs. Mealy Machine

Moore Machine: $Z = \lambda(S)$



Mealy Machine: $Z = \lambda(S, X)$



4.2 State Reduction and State Assignment

- **Equivalent States:** Two states S_1 and S_2 are equivalent if for every possible input sequence, they produce identical outputs and transition to identical (or equivalent) next states.
- **State Reduction:** Performed using row elimination or the **Implication Table method** to eliminate redundant states, directly minimizing flip-flop count.
- **State Assignment Techniques:**
  - **Binary Encoding:** Uses minimum number of flip-flops ($n = \lceil \log_2 N \rceil$), requires more combinational decode logic.
  - **Gray Encoding:** Consecutive states differ by 1 bit, minimizing switching noise and power.
  - **One-Hot Encoding:** Uses 1 flip-flop per state (N flip-flops for N states). Maximizes speed and simplifies next-state decode logic; standard choice in FPGA designs.

5. Design of Synchronous Circuits

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5.1 Sequence Detector / Pulse Train Generator

A sequence detector inspects an incoming serial binary bit stream and asserts output $Z = 1$ whenever a specific target pattern (e.g., 1011) is recognized.

- **Overlapping Sequence Detection:** The final bit(s) of a detected sequence can serve as the prefix of the next sequence. For 1011 with input 1011011 ..., two detections occur.
- **Non-Overlapping Sequence Detection:** The detector resets to the initial state upon finding the sequence; subsequent detection starts afresh.

5.2 Pseudo-Random Binary Sequence (PRBS) Generator

A PRBS generator produces a deterministic, periodic binary sequence possessing noise-like statistical properties using a **Linear Feedback Shift Register (LFSR)**.

- Constructed from an n-stage shift register where the input to the first flip-flop is the XOR sum of specific tap outputs defined by a primitive polynomial:

$$P(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_1 X + 1 \tag{8}$$

- **Maximal-Length Sequence (m-sequence):** When $P(X)$ is primitive, the LFSR cycles through all possible non-zero states:

$$L = 2^n - 1 \tag{9}$$

  (The all-zero state 00 ... 0 must be avoided because $\text{XOR}(0, 0) = 0$, causing lock-up).

- **Key Applications:** Spread-spectrum communications (CDMA PN codes), cryptographic stream ciphers, and Built-In Self-Test (BIST) pseudo-random test pattern generators.

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6. Quick Revision Page

Key Sequential Formulas and Equations at a Glance

- **SR Flip-Flop Equation:** $Q_{n+1} = S + \overline{R}Q_n \quad (S \cdot R = 0)$
- **JK Flip-Flop Equation:** $Q_{n+1} = J\overline{Q}_n + \overline{K}Q_n$
- **D Flip-Flop Equation:** $Q_{n+1} = D$
- **T Flip-Flop Equation:** $Q_{n+1} = T \oplus Q_n$
- **Race-Around Condition:** $t_{pd} < t_w < T/2 \quad (J = K = 1)$
- **Ripple Counter Maximum Delay:** $t_{\text{total}} = n \cdot t_{pd} \implies f_{\max} = 1/(n \cdot t_{pd})$
- **Ring Counter States:** $N = n \quad (n\text{-bit shift register})$
- **Johnson Counter States:** $N = 2n \quad (n\text{-bit shift register})$
- **Maximal LFSR / PRBS Period:** $L = 2^n - 1$

One-Line Definitions

- **Setup Time (t_s):** Minimum interval data must remain stable before the active clock edge.
- **Hold Time (t_h):** Minimum interval data must remain stable after the active clock edge.
- **Race-Around Condition:** Repeated uncontrolled toggling of a level-triggered JK flip-flop during clock high when $t_{pd} < t_w$.
- **Moore Machine:** A finite state machine whose output is strictly a function of current state only.
- **Mealy Machine:** A finite state machine whose output depends simultaneously on current state and current inputs.
- **Johnson Counter:** An inverted feedback shift register producing $2n$ states for n flip-flops.

Frequently Asked / Viva Questions

Q1. What is the difference between a latch and a flip-flop?

A latch is level-triggered (transparent whenever clock/enable is active), whereas a flip-flop is edge-triggered (samples and locks data strictly on clock transition edges).

Q2. Explain the race-around condition in a JK flip-flop. How is it eliminated?

When $J = K = 1$ in a level-triggered JK FF, if pulse width $t_w > t_{pd}$, the output toggles multiple times before the clock drops, leaving an uncertain final state. It is eliminated by using edge triggering or a Master-Slave configuration.

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Frequently Asked / Viva Questions (contd.)

Q3. Why is Master-Slave JK flip-flop pulse-triggered rather than edge-triggered?

Because the master accepts data while the clock is high, and data transfers to the slave on the falling edge; output changes only once per complete clock period.

Q4. How do you convert a JK flip-flop into a D flip-flop?

Connect input D directly to J, and connect an inverted D ($\overline{D}$) to K ($J = D, K = \overline{D}$).

Q5. What is the propagation delay limitation of ripple counters?

Carries must ripple through each stage sequentially ($t = n \cdot t_{pd}$), which limits the maximum clock frequency and creates transient false decoding spikes (glitches).

Q6. Compare Ring counter and Johnson counter for an n-bit shift register.

A ring counter yields n states and requires no decoding gates. A Johnson counter yields $2n$ states and requires simple 2-input AND gates for decoding.

Q7. What are the key architectural differences between Mealy and Moore FSMs?

In a Moore machine, output depends solely on current state ($Z = \lambda(S)$); in a Mealy machine, output depends on both current state and inputs ($Z = \lambda(S, X)$).

Q8. Why does a Mealy machine typically require fewer states than a Moore machine?

Because Mealy outputs can vary immediately within a state based on different inputs, whereas a Moore machine requires distinct states for distinct output combinations.

Q9. What is meant by One-Hot state assignment?

An encoding scheme using N flip-flops for N states where exactly one flip-flop is high ('1') at any time. It minimizes combinational decoding delay and is optimal for FPGAs.

Q10. What is an LFSR and why can the all-zero state not be used?

A Linear Feedback Shift Register uses XOR taps to generate pseudo-random sequences. If all flip-flops become 0, $\text{XOR}(0, 0) = 0$, locking the register in permanent zeros.

Q11. What is metastability in digital circuits?

A condition where a flip-flop's output hovers at an undefined intermediate voltage level between logic 0 and 1 for an indeterminate duration due to a setup/hold time violation.

Q12. How does an overlapping sequence detector differ from a non-overlapping one?

An overlapping detector allows the trailing bits of a detected sequence to be reused as the leading bits of the subsequent sequence, producing more output pulses.

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Comparison: Ripple vs. Synchronous Counters and Mealy vs. Moore FSM

Category	Architecture A	Architecture B	Key Design Trade-off
Counter Clocking	**Ripple Counter:** Flip-flops clocked by previous output	**Synchronous Counter:** All flip-flops clocked simultaneously	Ripple uses simpler wiring but has high cumulative delay; Synchronous achieves maximum speed.
FSM Output Logic	**Moore Machine:** $Z = \lambda(S)$ (State only)	**Mealy Machine:** $Z = \lambda(S, X)$ (State & Input)	Moore is glitch-free and simpler; Mealy responds 1 cycle faster with fewer states.
Shift Register Counting	**Ring Counter:** $N = n$ states, uncomplemented loop	**Johnson Counter:** $N = 2n$ states, inverted loop	Johnson provides double the count states with minimal decoding hardware.

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